

CT Reconstruction Software – Environment Variables (with source files)

Based on a getenv cross-scan of all 110 files in xrm-bl/ct-rec src/ / 2026-06-20

Thread / parallelism control				
Variable	Purpose	Default	Source files (src/)	Notes
CBP_THREADS	Worker threads for the CPU CBP (back-projection)	8	cbp_thread.c cbp_thread_int.c cbp_thread_nai.c	Outside 1..M (number of projections) -> bad number of CBP_THREADS
THREADS	MSD-comparison threads for rotation-axis xy / offset search	40 / 8 (see notes)	oct_xy.c (40) otf_xy.c (40) ofct_D0.c (8)	Default differs by program (oct_xy / otf_xy = 40, ofct_DO = 8). Value <= 0 -> bad number of THREADS
OMP_NUM_THREADS	Standard OpenMP variable; thread count for the ring-removal sort filter	logical cores (OpenMP default)	sort_filter_omp.c sort_filter_g.cu	Read by get_num_threads_from_env(). Value <= 0 is invalid -> default

Note: the GPU sort uses an internal SORT_THREADS=256; GPU filters use BLOCK_SIZE_X/Y/Z=8/8/4 (compile-time constants, not environment variables).

Ring removal				
Variable	Purpose	Default	Source files (src/)	Notes
KERNEL_SIZE	Kernel width of the sort / median filter for ring-artifact removal	5	sort_filter_omp.c sort_filter_g.cu	Current-repo default is 5 (changed from 3). Validity check is size > -1 (accepts >= 0)

Reconstruction quality				
Variable	Purpose	Default	Source files (src/)	Notes
CT_REC_BLACK_THRESHOLD	Black (dark) projection threshold. If the mean signal id_sum/N is below it, the frame is treated as a black projection and corrected	1.0	ct_rec.c otf_rec.c tf_rec.c rhp.c	Referenced in 4 files (rhp.c is the shared reader). getenv -> atof, default 1.0

Memory-aware chunking (common to 3 programs)				
Variable	Purpose	Default	Source files (src/)	Notes
HPTG_MEM_FRACTION	Fraction of available memory the program may use (for band sizing)	0.8	hp_tg_ku.c ofct_srec.c p_rec.c	Valid range $0.05 < v \leq 0.95$; outside the range the default 0.8 is kept
HPTG_MEM_LIMIT_MB	Absolute memory cap (MB)	unset = fraction-based	hp_tg_ku.c ofct_srec.c p_rec.c	If mb > 0, the budget is replaced by mb * 1024 * 1024 bytes (absolute cap)
HPTG_CHUNK_ROWS	Manual rows-per-band override	unset = auto	hp_tg_ku.c ofct_srec.c p_rec.c	If atoi > 0, forced to that many rows (clamped to 1..total_rows); otherwise auto = budget / bytes_per_row

Additional environment variables found (outside the original list, for reference)

Variable	Purpose	Default	Source files (src/)	Notes
CUDA_GPU	CUDA device index to use	0	<code>cu.h</code> (SETUP_CUDA_GPU)	Used by the GPU programs. Negative / invalid -> "no device assigned"
RHP_O	Override the path of the output.log to read	output.log	<code>rhp.c</code> <code>rtf.c</code>	HiPic / TIFF shared reader
RHP_D	Override the dark-image file path	dark.img / dark.tif	<code>rhp.c</code> <code>rtf.c</code>	same
RHP_Q	Override the projection-file prefix letter	q	<code>rhp.c</code> <code>rtf.c</code>	Uses the first character (a-z); e.g. changes the "q" in <code>q[0-9]*.img</code>

Method: `git clone` then `grep -rn getenv src/*.c src/*.cu src/*.h` across all 110 files. The variables above are every env var read via `getenv` (including `CUDA_GPU` / `RHP_*`). The `KERNEL_SIZE` in `tif_blf_g.cu` is a macro (`MAX_KERNEL_SIZE`), not an environment variable, so it is excluded.